

Test-Driven

Development

Yegor Bugayenko

Lecture #13 out of 16
80 minutes

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The Psychology of Testing

Test Driven Development (TDD)

Unit vs. Integration Tests

Test Coverage

Automated Performance Testing

Behavior Driven Development (BDD)

Testing vs. QA

Books, Venues, Call-to-Action

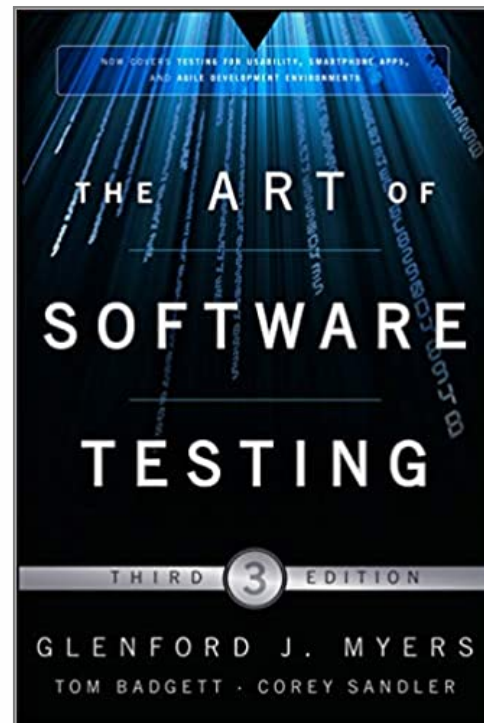
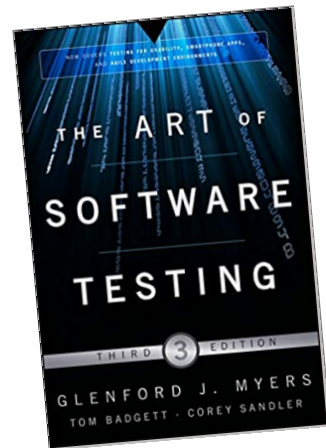
Chapter #1:

The Psychology of Testing



“One of the primary causes of poor application testing is the fact that most programmers begin with a false definition of the term. They might say: ‘Testing is the process of demonstrating that errors are not present.’”

— Glenford J. Myers, [The Art of Software Testing](#)



“Don’t test a program to show that it works; rather, start with the assumption that the program contains errors. Testing is the process of executing a program with the intent of finding errors.”

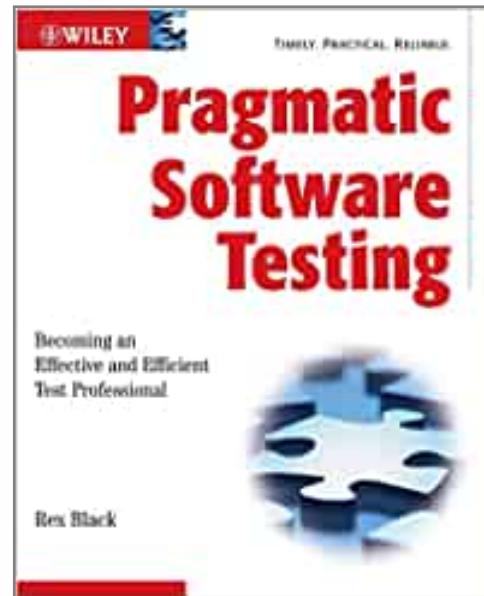
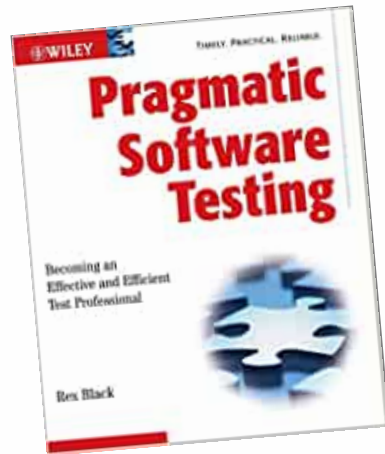
— Glenford J. Myers et al. *The Art of Software Testing*. 2nd ed. Wiley, 2012.
DOI: [10.5555/2161638](https://doi.org/10.5555/2161638)

“If something is to be delivered, then it is the testers who make the final decision as to whether or not that something is delivered into the live environment.” —Nick Sewell, *How to Test a System That Is Never Finished*, 2009

“Testing is an essential activity in software engineering. In the simplest terms, it amounts to observing the execution of a software system to validate whether it behaves as intended ...” —Antonia Bertolino, *Software testing research: Achievements, challenges, dreams*, 2007

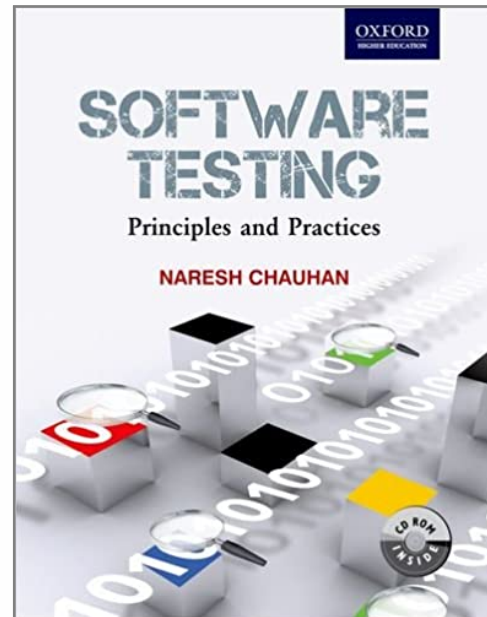
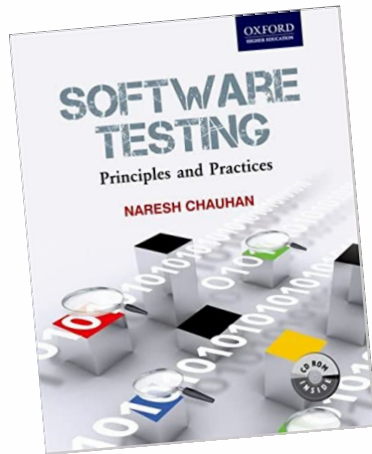
“Software testing is the process of executing a software system to determine whether it matches its specification and executes in its intended environment.” —James A. Whittaker, *What Is Software Testing? And Why Is It So Hard?*, 2000

“We distinguish the four major testing models... One model says we test to demonstrate that some version of the software satisfies its specification, two models say we test to detect faults, and the fourth says we test to prevent faults. These three goals need not conflict and, in fact, are all present in the prevention model.” —David Gelperin, *The growth of software testing*, 1988



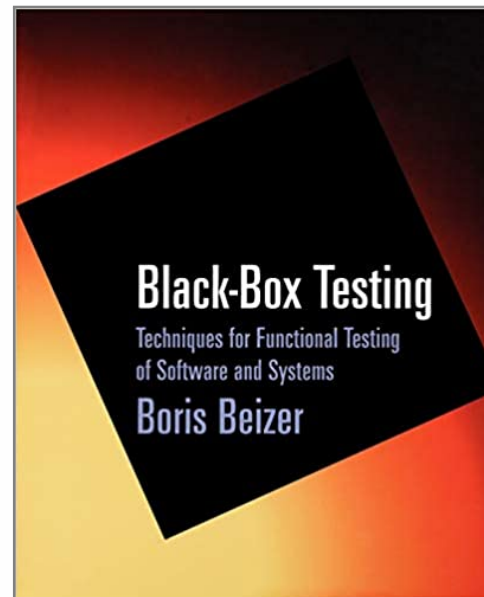
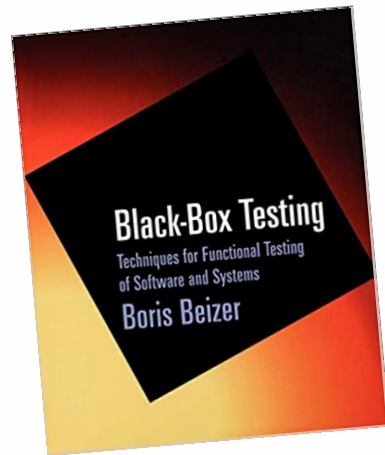
“Software testing is not about proving conclusively that the software is free from any defects, or even about discovering all the defects. Such a mission for a test team is truly impossible to achieve.”

— Rex Black. *Pragmatic Software Testing: Becoming an Effective and Efficient Test Professional*. John Wiley & Sons, 2007. DOI: 10.5555/1215210



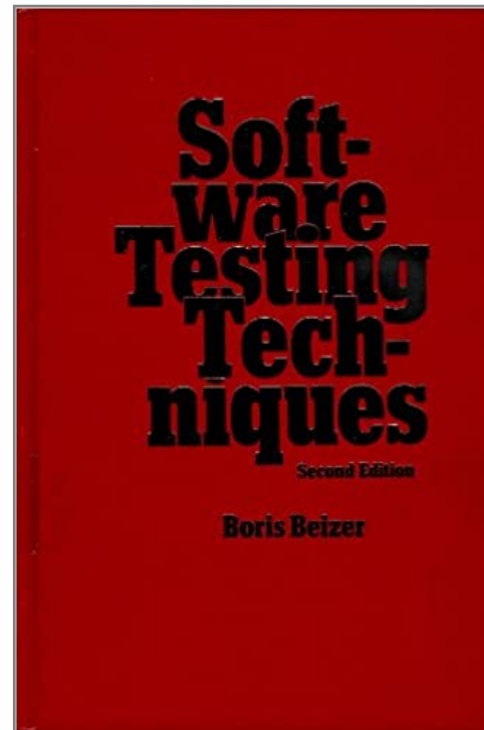
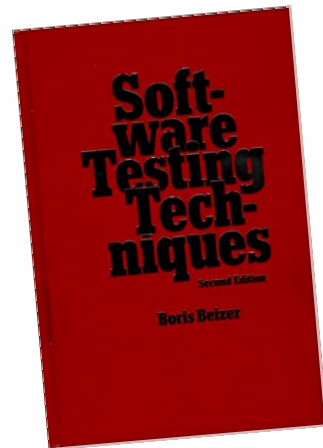
“The immediate goal of testing is to find errors at any stage of software development. More the bugs discovered at an early stage, better will be the success rate of software testing.”

— Naresh Chauhan. *Software Testing: Principles and Practices*. Oxford University Press, 2010



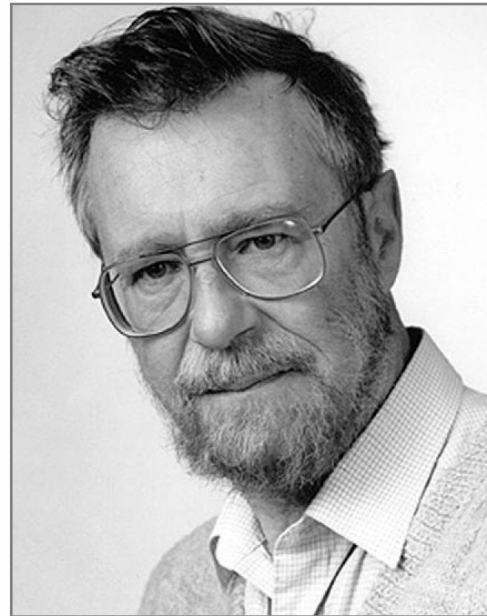
“Anything written by people has bugs. Not testing something is equivalent to asserting that it’s bug-free. Programmers can’t think of everything especially of all the possible interactions between features and between different pieces of software. We try to break software because that’s the only practical way we know of to be confident about the product’s fitness for use.”

— Boris Beizer. *Black-Box Testing: Techniques for Functional Testing of Software and Systems*. John Wiley & Sons, 1995. DOI: [10.5555/202699](https://doi.org/10.5555/202699)



“The probability of showing that the software works decreases as testing increases; that is, the more you test, the likelier you are to find a bug. Therefore, if your objective is to demonstrate a high probability of working, that objective is best achieved by not testing at all!”

— Boris Beizer. *Software Testing Techniques*. 2nd ed. Van Nostrand Reinhold, 1990. DOI: [10.5555/79060](https://doi.org/10.5555/79060)



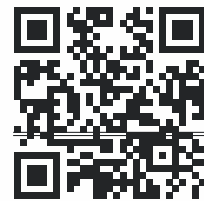
“Program testing can be used to show the presence of bugs, but never to show their absence!”

— Edsger W. Dijkstra. *Notes on Structured Programming*. 1970

Software Testing Philosophy

Bee Mobile Meetup

Moscow, Russia, 7 November 2018



Software Testing Pitfalls

JPoint

Moscow, Russia, 5 April 2019



Testing and Testers

TestCon

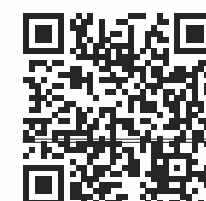
Moscow, Russia, 16 September 2020



Quality Assurance vs. Testing

QA Fest

Kyiv, Ukraine, 20 September 2019



Chapter #2:

Test Driven Development (TDD)

```
1 (test one
2   (is (= 1 (f 1))))
3 (test two
4   (is (= 1 (f 2))))
5 (test fifteen
6   (is (= 610 (f 15))))
```

Can you put some code here?

```
1 | (test one
2 |   (is (= 1 (f 1))))
3 | (test two
4 |   (is (= 1 (f 2))))
5 | (test fifteen
6 |   (is (= 610 (f 15))))
```

```
1 | (defun f (n)
2 |   (cond
3 |     ((= n 1) 0)
4 |     ((= n 2) 1)
5 |     (+
6 |       (f (- n 1))
7 |       (f (- n 2)))))
```



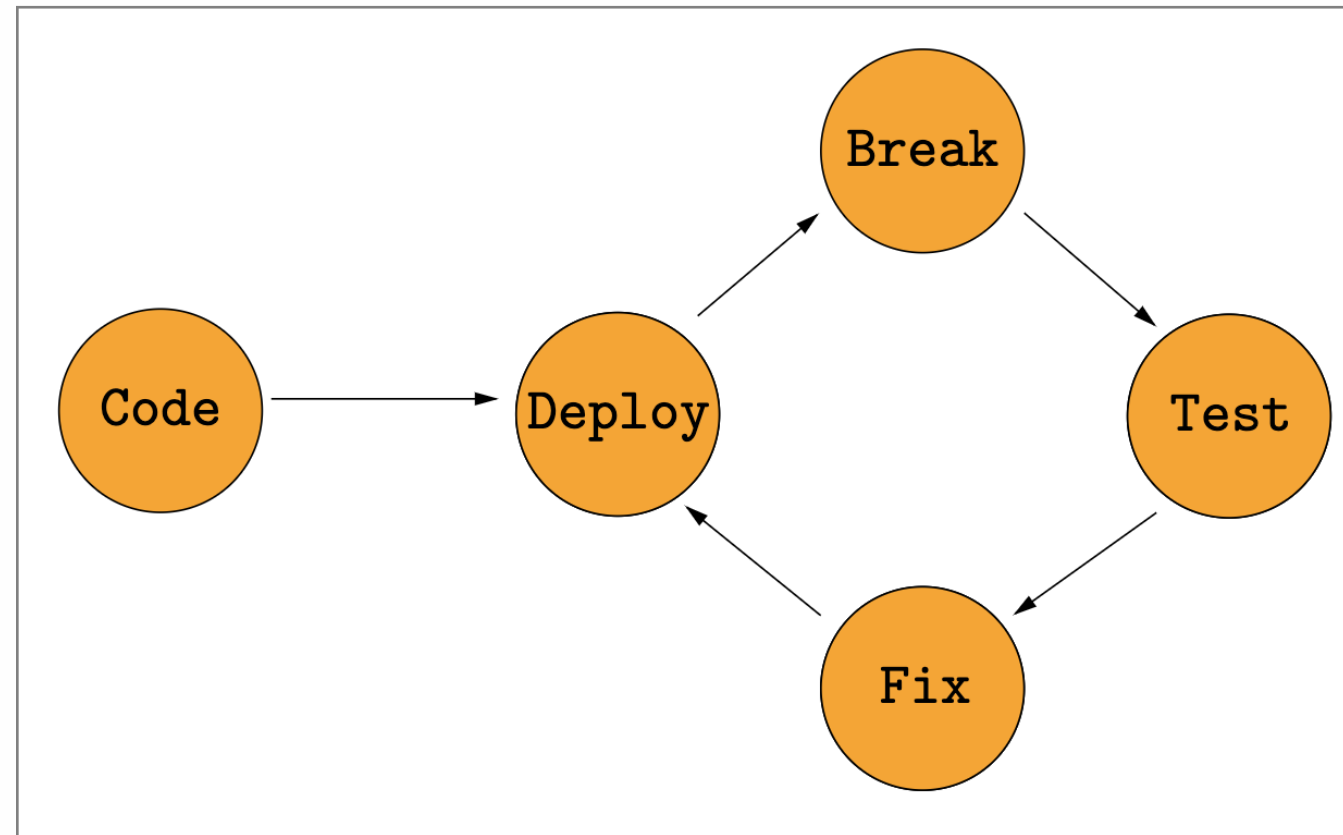
“Test-first fundamentalism is like abstinence-only sex ed: An unrealistic, ineffective morality campaign for self-loathing and shaming.”

— David H. Hansson. *TDD Is Dead. Long Live Testing.*
<https://dhh.dk/2014/tdd-is-dead-long-live-testing.html>. [Online; accessed 04-05-2025]. 2014



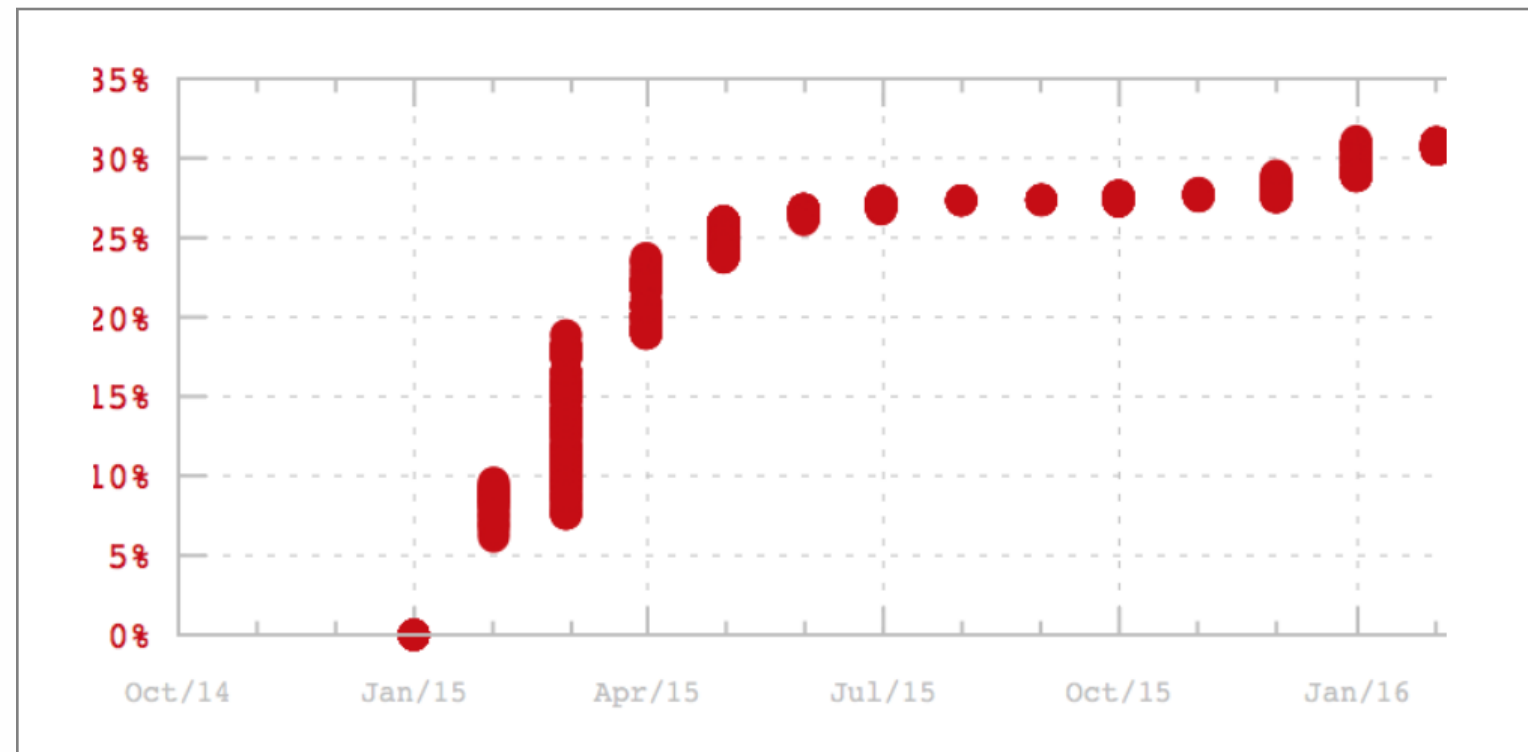
“It would not surprise me if, one day, TDD had the force of law behind it.”

— Robert C. Martin. *Professionalism and TDD (Reprise)*.
<https://shorturl.at/nGS3G>. [Online; accessed 04-05-2025]. 2014



“I only create tests later when my users express the need for them by reporting bugs.”

<https://www.yegor256.com/2017/03/24/tdd-that-works.html>



“ I don't need tests at the beginning of the project”

<https://www.yegor256.com/2017/03/24/tdd-that-works.html>

Safety Net

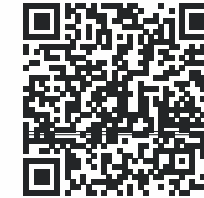


Chapter #3:

Unit vs. Integration Tests

Good Tests Are:

- 1) **Short**: less than x lines each
- 2) **Fast**: less than y milliseconds each
- 3) **Independent**: runs alone and in a suite
- 4) **Portable**: runs on your laptop and on mine
- 5) **Careful**: side effect free, doesn't leave temp files
- 6) **Isolated**: doesn't touch my files



<https://www.kenneth-truvers.net/2012/12/15/key-qualities-of-a-good-unit-test/> ➞

```
1 import java.nio.file.Files;
2 class Book {
3     String title() {
4         return Files.readAllLines(
5             Paths.get("/my-data/book.txt
6         ")
7         )[0];
8     }
9 }
```

```
1 import org.junit.jupiter.api.
   Assertions;
2 import org.junit.jupiter.api.Test;
3 class BookTest {
4     @Test
5     void canRetrieveTitle() {
6         String t = new Book().title();
7         Assertions.assertEquals(
8             "Object Thinking", t
9         );
10    }
11 }
```

Is it Short, Fast, Independent, Portable, Careful, and Isolated?

```
1 import java.nio.file.Files;
2 class Book {
3     private Path file;
4     Book(Path f) {
5         this.file = f;
6     }
7     String title() {
8         return Files.readAllLines(
9             this.file
10            )[0];
11     }
12 }
```

```
1 import org.junit.jupiter.api.
    Assertions;
2 import org.junit.jupiter.api.Test;
3 class BookTest {
4     @Test
5     void canRetrieveTitle() {
6         Path f = Paths.get("/tmp/temp.
7            txt");
8         String title = "Object Thinking
9            ";
10        Files.write(f, title.getBytes());
11        String t = new Book().title();
12        Assertions.assertEquals(title,
13            t);
14    }
15 }
```

11 | }

12 | }

Is it Short, Fast, Independent, Portable, Careful, and Isolated?

```
1 import java.nio.file.Files;
2 class Book {
3     private Path file;
4     Book(Path f) {
5         this.file = f;
6     }
7     String title() {
8         return Files.readAllLines(
9             this.file
10            )[0];
11     }
12 }
```

```
1 import org.junit.jupiter.api.
    Assertions;
2 import org.junit.jupiter.api.Test;
3 import org.junit.jupiter.api.io.
    TempDir;
4 class BookTest {
5     @Test
6     void canRead(@TempDir Path dir) {
7         Path f = dir.resolve("temp.txt
8         ");
9         String title = "Object Thinking
10        ";
11         Files.write(f, title.getBytes()
12        );
13         String t = new Book().title();
```

```
11 | Assertions.assertEquals(title,  
    | t);  
12 | }  
13 | }
```

Chapter #4:

Test Coverage

```
195   loop do
196     json = @api.block(block)
197     if json[:orphan]
198       steps = 4
199       @log.info("Orphan block found at #{block}, moving #{steps} steps back...")
200       wrong << block
201       steps.times do
202         block = json[:previous]
203         wrong << block
204         @log.info("Moved back to #{block}")
205         json = @api.block(block)
206       end
207     next
208   end
209   checked = 0
210   checked_outputs = 0
211   json[:txns].each do |t|
212     t[:outputs].each_with_index do |o, i|
213       address = o[:address]
214       checked_outputs += 1
```

<https://codecov.io/gh/yegor256/sibit/tree/master/lib>

Coverage Criteria

Function Coverage

Has each function (or subroutine) in the program been called?

Statement Coverage

Has each statement in the program been executed?

Edge Coverage

Has every edge in the control-flow graph been executed?

Branch Coverage

Has each branch of each control structure been executed?

Condition Coverage

Has each Boolean sub-expression evaluated both to true and false?

Mutation Testing + Coverage

```
1 | def f(n)  
2 |   n * n + 1  
3 | end
```

```
1 | f(0) == 1  
2 | f(1) == 2  
3 | f(2) == 5
```

Mutation Testing + Coverage

```
1 def f(n)
2   n * n + 1
3 end
4
5 # Mutant no. 1
6 def f(n)
7   n + n + 1
8 end
9
10 # Mutant no. 2
11 def f(n)
12   n * n - 1
13 end
```

```
1 f(0) == 1
2 f(1) == 2
3 f(2) == 5
```

Test coverage: 100%

Mutation coverage = 50%

Chapter #5:

Automated Performance Testing

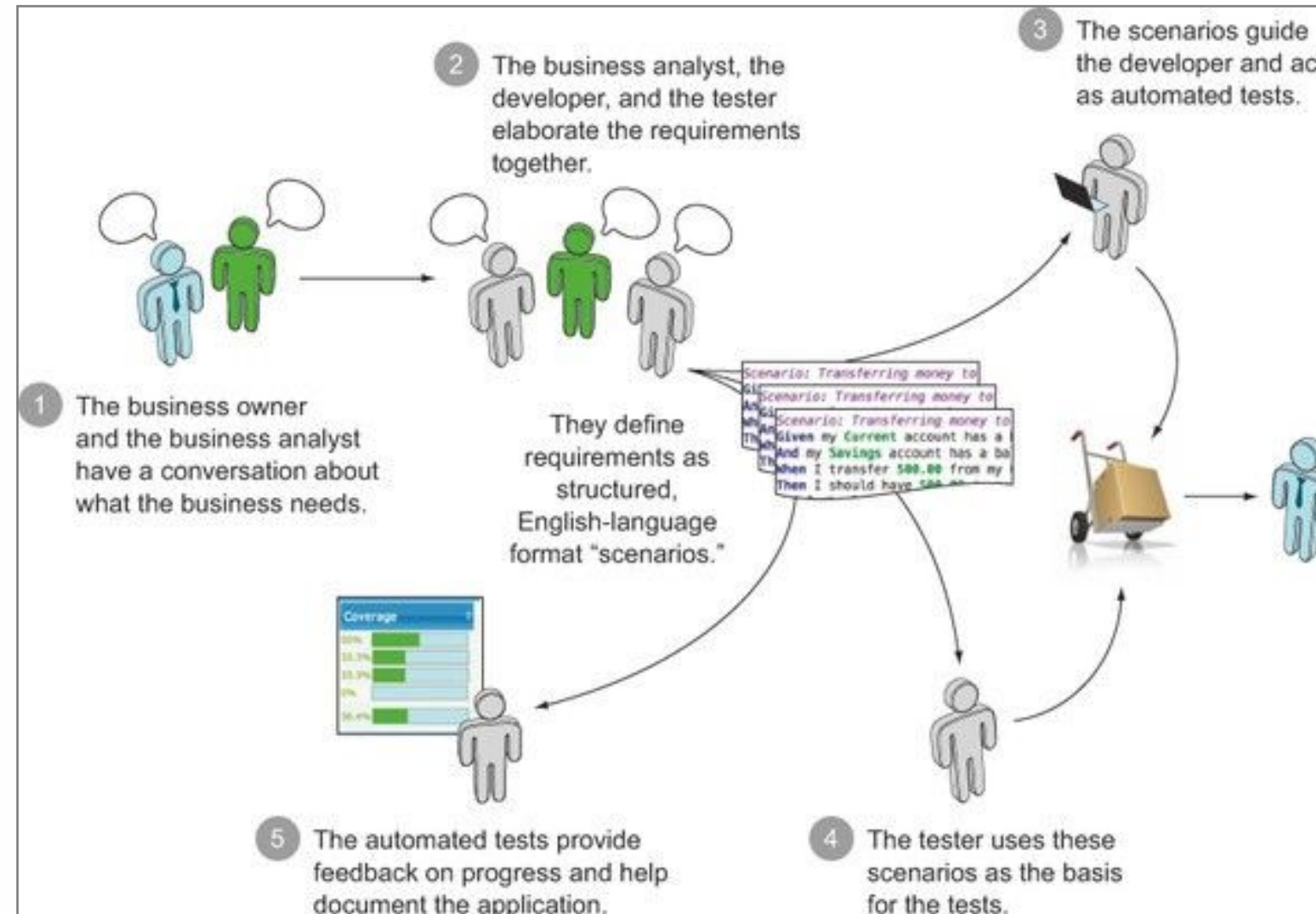
Some Tools

WebLOAD
LoadNinja
ReadyAPI Performance
LoadView
StormForge
Keysight's Eggplant
Apache JMeter
LoadRunner
Appvance

NeoLoad
LoadComplete
WAPT
Loadster
k6
Rational Performance Tester
Testing Anywhere
Apache Bench

Chapter #6:

Behavior Driven Development (BDD)



<https://www.agilealliance.org/glossary/bdd>

Chapter #7:

Testing vs. QA



<https://www.youtube.com/watch?v=jZitXMQaXvE> →

Testing $\neq$ Quality Assurance (QA)

QA

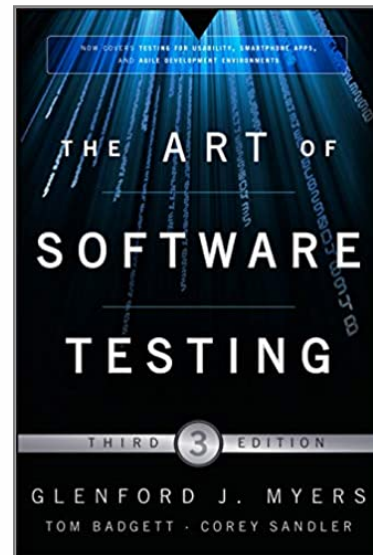
Quality Assurance is the process of auditing the quality requirements and the results from quality control measurements to ensure that appropriate quality standards and operational definitions are used.

QC

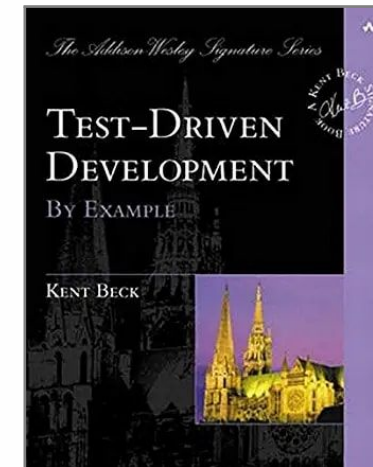
Quality Control is the process of monitoring and recording results of executing the quality activities to assess performance and recommend necessary changes.

Chapter #8:

Books, Venues, Call-to-Action



Glenford J. Myers et al. *The Art of Software Testing*.
2nd ed. Wiley, 2012. DOI: [10.5555/2161638](https://doi.org/10.5555/2161638)



Kent Beck. *Test Driven Development: By Example*.
Addison-Wesley, 2002. DOI: [10.5555/579193](https://doi.org/10.5555/579193)

Where to publish:

International Symposium on Software Testing and Analysis (ISSTA)

Call to Action:

Integrate mutation coverage control into your build.

Still unresolved issues:

- How to test performance right?
- How to create tests automatically?
- How to motivate programmers write tests?
- How to control the quality of testing?

References

- [1] Kent Beck. *Test Driven Development: By Example*. Addison-Wesley, 2002. DOI: [10.5555/579193](https://doi.org/10.5555/579193).
- [2] Boris Beizer. *Black-Box Testing: Techniques for Functional Testing of Software and Systems*. John Wiley & Sons, 1995. DOI: [10.5555/202699](https://doi.org/10.5555/202699).
- [3] Boris Beizer. *Software Testing Techniques*. 2nd ed. Van Nostrand Reinhold, 1990. DOI: [10.5555/79060](https://doi.org/10.5555/79060).
- [4] Rex Black. *Pragmatic Software Testing: Becoming an Effective and Efficient Test Professional*. John Wiley & Sons, 2007. DOI: [10.5555/1215210](https://doi.org/10.5555/1215210).
- [5] Naresh Chauhan. *Software Testing: Principles and Practices*. Oxford University Press, 2010.
- [6] Edsger W. Dijkstra. *Notes on Structured Programming*. 1970.
- [7] David H. Hansson. *TDD Is Dead. Long Live Testing*. <https://dhh.dk/2014/tdd-is-dead-long-live-testing.html>. [Online; accessed 04-05-2025]. 2014.
- [8] Robert C. Martin. *Professionalism and TDD (Reprise)*. <https://shorturl.at/nGS3G>. [Online; accessed 04-05-2025]. 2014.
- [9] Glenford J. Myers et al. *The Art of Software Testing*. 2nd ed. Wiley, 2012. DOI: [10.5555/2161638](https://doi.org/10.5555/2161638).